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DVMDOSTEM: a terrestrial ecosystem model designed to represent arctic, boreal and permafrost ecosystem dynamics

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Software

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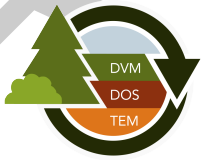

Figure 1: Logo for DVMDOSTEM

Summary

The impacts of climate change on natural ecosystems are the result of complex physical and ecological processes operating and interacting at a variety of spatio-temporal scales. For this reason, process-based ecosystem models are efficient tools to formalize and extrapolate our current understanding of ecosystem dynamics and predict local, regional and global climate impacts. As such, process-based models have the potential to provide critical information for adaptation and mitigation efforts around the globe.

DVMDOSTEM is an advanced process-based terrestrial ecosystem model (TEM) designed to study ecosystem responses to climate changes and disturbances. It has a particular focus on permafrost regions (i.e. regions characterized by soils that stay partially frozen all year round for at least two consecutive years), encompassing boreal, arctic, and alpine landscapes. The model couples a dynamic vegetation module (DVM) and a dynamic organic soil (DOS) module to simulate processes at yearly and monthly scales, with some physical processes operating at an even finer temporal resolution. Its versatility allows for site-specific to regional simulations, making it valuable for predicting shifts in permafrost, vegetation, and carbon (C) and nitrogen (N) dynamics across arctic, boreal, and alpine regions.

Statement of need

Arctic and boreal regions underlain by permafrost store nearly half of the world's soil organic C - approximately 1,440-1,600 Pg (Hugelius et al., 2014; Edward A. G. Schuur et al., 2022). These regions are warming roughly two to four times faster than the rest of the globe, driving widespread and rapid permafrost thaw (Rantanen et al., 2022; Smith et al., 2022). As permafrost thaws, soil organic C becomes available for decomposition and release as greenhouse gasses (GHGs) to the atmosphere. Climate-driven permafrost thaw and the associated release of GHGs can influence the global climate system, a phenomenon called the permafrost carbon-climate feedback or PCCF (Koven et al., 2011; E. A. G. Schuur et al., 2015). The PCCF has been identified as one of the largest sources of uncertainty in future climate projections and therefore needs to be accurately represented in process-based ecosystem models that inform global earth system models (Schädel et al., 2024).

Model Design

`DVMDOSTEM` is a process-based ecosystem model designed to simulate the key biophysical and biogeochemical processes between the soil, the vegetation and the atmosphere. The evolution and refinement of `DVMDOSTEM` have been shaped by extensive research programs and applications both in permafrost and non-permafrost regions (Briones et al., 2024; Eugénie S. Euskirchen et al., 2022; H. Genet et al., 2013; Hélène Genet et al., 2018; E. E. Jafarov et al., 2013; Yi et al., 2009, 2010). The model is spatially explicit and represents ecosystem response to climate and disturbances at seasonal (i.e. monthly) up to centennial scales. The snow and soil columns are split into a dynamic number of layers to represent their impact on thermal and hydrological dynamics and the consequences for soil C and N dynamics. Vegetation composition is modeled using community types (CMTs), each of which consists of multiple plant functional types (PFTs - groups of species sharing similar ecological traits). This structure allows the model to represent the effect of competition for light, water and nutrients on vegetation composition (E. S. Euskirchen et al., 2009). The model also represents the ecosystem N cycle to evaluate the role of nutrient limitations, characteristic of permafrost regions, on ecosystem dynamics, with coupling between the C and N cycles (E. S. Euskirchen et al., 2009; A. D. McGuire et al., 1992). Finally, the model represents the effects of wildfire in order to evaluate the role of climate-driven fire intensification on ecosystem structure and functions (H. Genet et al., 2013; Yi et al., 2010). The structure of `DVMDOSTEM` is represented visually in Figure 2.

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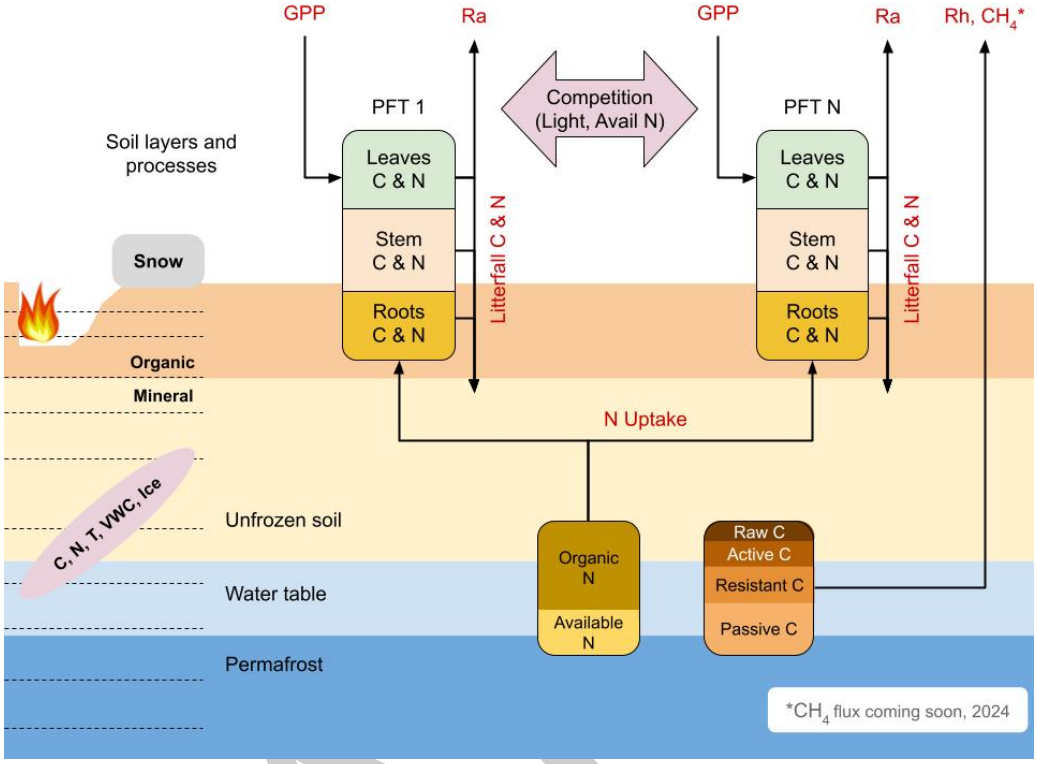

Figure 2: Overview of DVMDOSTEM soil and vegetation structure. On the left is the soil structure showing the layers and different properties that are tracked (purple bubble: carbon (C), nitrogen (N), temperature (T), volumetric water content (VWC), ice). Each of the layers with properties described above is also categorized as organic (fibric or humic) or mineral. Additionally, the model simulates snow layers and the removal of soil organic layers due to processes such as fire. On the right is the vegetation structure showing the plant functional types (PFTs) within a given community type (CMT) and the associated pools and fluxes of C and N. Each PFT is split into compartments (leaf, stem and root) which track their own C and N content and associated fluxes. The fluxes are represented with red text while the pools are black. In addition, there is competition among the PFTs for light, water, and available N, shown with the purple arrow in the top center.

Snow and soil structure and processes

The soil column is structured as a sequence of layers organized by soil horizons (i.e. fibric and humic organic horizons, mineral horizon and parent material). The number and physical properties of layers may change throughout the simulation based on vegetation, thermal, hydrologic, and seasonal properties that are calculated at each time step (Eugénie S. Euskirchen et al., 2014; A. David McGuire et al., 2018; Yi et al., 2009; Zhuang et al., 2003). The model uses the two-directional Stefan algorithm and the Richards equation to predict freezing/thawing fronts and soil moisture dynamics in the unfrozen layers, respectively (Yi et al., 2009, 2010; Zhuang et al., 2003). Snow is also represented with a dynamic stack of layers. The physical properties of the snowpack (density, thickness, and temperature) are calculated from snowfall, sublimation and snowmelt. The snow cover influences soil-thermal and hydrological seasonal dynamics. Changes in soil C stocks are a result of litterfall from the vegetation and decomposition of soil C stocks by microbes (heterotrophic respiration or Rh). Changes in soil organic and available N stocks are a result of litterfall, net mineralization of organic N, and plant N uptake. Soil organic layers and soil C and N stocks may also be modified as a result of wildfire.

Vegetation structure and processes

Each vegetation CMT (e.g. "wet-sedge tundra", "white spruce forest", etc.), is modeled with up to ten PFTs (e.g., "deciduous shrubs", "sedges", "mosses", etc.), each of which may have up to three compartments: leaf, stem, and root. Vegetation C and N fluxes are calculated at each time step based on environmental factors and soil properties. Assimilation of atmospheric CO_2 by the vegetation is estimated by computing gross primary productivity (GPP) for each PFT. This C flux is a function of foliage development (seasonal and successional patterns), air and soil temperature, water and nutrient availability, photosynthetically active radiation, and maximum assimilation rate (a calibrated parameter) (E. S. Euskirchen et al., 2009; A. D. McGuire et al., 1992). Changes in vegetation C stocks are calculated using GPP, autotrophic respiration (Ra), and litterfall (transfer from vegetation to soil). Vegetation N stocks are calculated using plant N uptake and litterfall. Vegetation C and N stocks may also be modified as a result of wildfire burn.

Run stages

To initialize an historical or future simulation, `DVMDOSTEM` needs to compute a quasi steady-state (QSS) solution. This solution is forced by using averaged historical atmospheric and ecosystem properties (e.g. soil texture) to drive the model. QSS of physical processes (e.g. soil temperature and water content) are usually achieved in less than 100 years, while QSS of biogeochemical processes (e.g. soil and vegetation C and N stocks) are achieved in 1,000 to >10,000 years. Additionally, biogeochemical QSS is achieved more rapidly if computed starting with QSS soil physical properties. To leverage this property for decreasing overall run-times, `DVMDOSTEM` uses two QSS stages: "Pre-run" and "Equilibrium". The list of all `DVMDOSTEM` run stages is as follows:

- Pre-run (pr): QSS computation for the physical state variables.
- Equilibrium (eq): QSS computation for the biogeochemical state variables.
- Spinup (sp): introduction of pre-industrial climate variability and fire regime.
- Transient (tr): historical simulation.
- Scenario (sc): future simulation.

A complete model simulation requires advancing the model consecutively through all of the run stages, however users are able to selectively skip stages in order to save compute time for certain debugging workflows or when running multiple scenarios.

Spatial considerations

`DVMDOSTEM` can be applied at the site level or across large regions. Spatially, `DVMDOSTEM` breaks up the landscape domain into grid cells, each of which is characterized by a set of input (forcing) values and a set of parameterization values. Parameterization values for a grid cell describe soil and vegetation characteristics and are associated with a CMT. DMVDOSTEM does not include the lateral transfer of information between grid cells (e.g. water, C, and nutrients). The CMT classification for each grid cell is static across the time dimension of a model simulation. These two factors limit the ability of the model to represent climate-driven biome shifts or succession trajectories from disturbances such as wildfire (Johnstone et al., 2010). Design discussions are in progress for adding these capabilities to `DVMDOSTEM`.

Inputs and outputs

`DVMDOSTEM` inputs and outputs are NetCDF files (Rew & Davis, 1990), which conform to the CF Conventions v1.11 (Eaton et al., 2011) where possible. The input variables used to drive `DVMDOSTEM` are: drainage classification (upland or lowland), CMT classification, topography (slope, aspect, elevation), soil texture (percent sand, silt, and clay), climate (air temperature, precipitation, vapor pressure, incoming shortwave radiation), atmospheric CO_2 concentration,

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and fire occurrence (date and severity). All input datasets are spatially explicit, except the time series of atmospheric CO_2.

There are approximately 110 different variables available for output from DVMDOSTEM. One file will be produced per requested output variable. Users can specify the temporal and structural resolutions at which model outputs are produced (e.g. yearly or monthly soil carbon stocks across the whole soil column or by layer). This functionality allows the users to consider their computational resources and information needs in the selection of output data to be produced.

Parameterization

DVMDOSTEM parameterization sets are developed for each CMT represented in the model. Each CMT is defined by more than 200 parameters. Parameter values are estimated directly from field, lab or remote sensing observations, literature review or site-specific calibration. Calibration is required when (1) parameter values cannot be determined directly from available data or published information, and (2) model sensitivity to the parameter is substantial. The calibration process consists of adjusting parameter values until there is acceptable agreement between measured field data and model prediction on the state variables most influenced by the parameter to be calibrated. Due to the large number of parameters requiring calibration, and the non-linear nature of the relationships between parameters and state variables, model calibration can be a labor intensive process. We are actively developing a calibration process that allows automation, leveraging parameter optimization methods to improve calibration performance (Elchin E. Jafarov et al., n.d.).

Software Design

The DVMDOSTEM software repository is a combination of several tightly coupled sub-components: - the DVMDOSTEM model itself, - supporting tools, and - a development environment specification.

At the core of the project is the DVMDOSTEM model itself. The DVMDOSTEM model is written in C++ and uses some object-oriented concepts. The model exposes a command line interface that allows users to start simulations individually or use a scripting language to drive the command line interface.

Surrounding the core model is a large body of supporting tooling to assist the user with preparing inputs, setting up and monitoring model runs and analyzing model outputs. This collection of tools is primarily written in Python and shell scripts, with some of the demonstration and exploratory analysis using Jupyter Notebooks. The supporting tooling is partially exposed via command line interfaces and a Python API which are documented in the User Guide.

The model and tools target a UNIX-like operating system environment. The combination of the core DVMDOSTEM model and the supporting tools result in the need for a complex computing environment with many dependencies. To manage this complexity and provide consistent environments for development and production, the repository provides definitions for Docker images (Merkel, 2014). The project includes the specification for several development images as well as a pared down runtime-only image.

The software design is imperfect - it is a work in progress and stems from the organic growth of the project spanning 30+ years of development by research scientists, graduate students and programmers. Recent years have seen an increased effort to apply professional software development practices such as version control, automated documentation, containerization, and testing.

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